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ROBUST STRUCTURED LIGHT PATTERN FOR USE WITH A SPATIAL LIGHT MODULATOR IN 3-D ENDOSCOPY

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This article introduces a novel structured light pattern designed to be compatible with the spatial light modulator (SLM) projection. The proposed pattern is a De Bruijn-based sequence applied to a combination of continuous and dashed lines for the pattern. The sequence is coded in the period and duty cycles of the dashed lines. It provides 16 different lines which limits to two the required number of dashed lines needed for identification. The segmentation has been made easier by alternating continuous and dashed lines. As required by the use of SLMs, the sequence has been adapted by making it symmetric. It has been improved by guaranteeing a hamming distance equal to two for two successive dashed lines. The implementation on a virtual model has shown that a subpixel accuracy has been achieved. This pattern has been developped for 3-D endoscopy.

Keywords: 3D reconstruction, 3D Endoscopy, diffractive optimal element pattern, spatial light modulator, structured light

1. INTRODUCTION

In the field of 3-D endoscopy, structured light is more frequently used. This technique provides a good way to solve the matching problem of the stereo vision. Some of them (Wu and Qu 2007; Albitar et al. 2007; Chen and Jie 2009) have proposed the use of a spatial light modulator (SLM) or a diffractive optical element for pattern projection. According to them, the latter can be easily embedded in an endoscope as it is minimally invasive. The authors have proposed patterns or devices using this technique. However, no proposed pattern compatible with the SLM projection provides a high resolution (Albitar et al. 2007) or is coded for robustness (Wu and Qu 2007). These issues are solved by the pattern proposed here. The second section introduces the patterns and the coding strategies adopted by the authors. The

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NOMENCLATURE

a	degree of the De Bruijn sequence	n	length of a De Bruijn sequence
$a_{i,j}$	Length of the segment in the line i at position j	h_i	Height of the segment$i$
d	Distance between two contiguous lines or distance between the camera and the projector	pi	Measuredperiod of the line i
		Pi	Calculated period of the line i
		SLM	Spatial light modulator
DOE	Diffractive optical element	$V_{i,j}$	Vector director if the segment in the line$i$ at position j
f_p	Focal length	Z	Working depth
FOV	Field of View	α,β	Angles between an observed point and camera's straight line
mis	the degree of the primitive polynomial used for De Bruijn sequence	$\rho(i)$	Autocorrelation of a sequence at position i

difference will be made between the sequence used to generate the pattern and the coding elements (called symbols here) used to draw the pattern. The third section introduces the pattern constructed for the application. The generated sequence is based on a De Bruijn sequence and the symbols are dashed lines. The last section details the experimentation on a model in order to quantify errors and the quality of the reconstruction.

2. STRUCTURED LIGHT AND PRIOR ART

2.1. Principle

The structured light technique is based on image capture of a known pattern projected on an unknown surface. The principle is well detailed in the tutorial made by Geng (2011). The position of the projector and of the camera combined with the analysis of the pattern deformation provides the required information to reconstruct the observed surface in 3-D. This technique solves the main weakness of the regular stereo vision: the match of the pixels in the left and right images. The pattern will, therefore, influence the robustness and the accuracy of the reconstruction. It has to be well designed to make the detection easier and to minimize the risk of confusion.

2.2. Spatial Light Modulator Projection

The SLM is a diffractive optical element used to project patterns. Its use in structured light is put forward by several authors. Some authors have proposed devices using it (Wu and Qu 2007; Albitar et al. 2007; Chen and Jie 2009; Freedman et al. 2008). The most important one is the device Prime Sense LTD (Tel-Aviv, Isreal) used in the Microsoft play console Kinect® (Freedman et al. 2008). The use of SLM brings several advantages such as the space occupied or its robustness. It also brings some constraints: it can only project a monochromatic pattern, and the pattern should have a central symmetry to maximize the transmitted power.

2.3. Pattern Coding Strategies

Many authors have defined their own pattern adapted to their applications. Salvi et al. (2010) have recently proposed a three-fold classification of the patterns described in the literature. They identify three main strategies for coding the patterns:

- Direct coding: every element is directly coded,
- Time multiplexing: works as the lighthouses, and
- Spatial neighborhood: items are defined thanks to their neighbors.

The methods based on the direct coding are the easiest to implement, as occlusions have no impact on the quality of the identification. However, most of the proposed patterns require either a full spectrum of colours or grey levels. They are, therefore, not adapted to colored objects, to a high number of different elements or to be used with a SLM. The methods based on time multiplexing achieve good results in terms of resolution, but they are not adapted to moving objects. Spatial neighborhood strategies are not the best in terms of spatial resolution, but they offer other advantages that make them well adapted to several situations. First, they are adapted to moving scenes as they remain unchanged in time. Second, they can easily be monochromatic and are, therefore, adapted to the SLM projection. Finally, they are less sensitive to light and color changes in the scene.

2.4. Defining the Sequence and the Symbols

Spatial neighborhood requires the identification of the neighborhood of an element to be able to identify the latter, just as a word is identified by its letters arranged in a certain way. It may be useful to remind some requirements. First, the smallest neighborhood (called the codeword) has to be unique for the match to be unique. Second, it should preferably be short; and third, the detection in image has to be made as easy as possible. These requirements can be fulfilled in different ways by optimizing the pattern made of symbols arranged by a sequence (generally, made of numbers). The next paragraphs present a prior art study of the symbols used to draw the pattern and of the sequence used to arrange them. Based on this, the next section introduces the pattern here introduced and solving the raised issues.

2.4.1. Sequence generating strategies. Here, are described three methods to build the sequence:

- The uncoded sequences,
- The nonformal sequences, and
- The De Bruijn-coded sequences.

Wu and Qu (2007) have proposed the use of a regular uncoded grid in their endoscopic application. However, this method may suffer from matching problems in case of occlusions and is therefore not robust. Beumier and (2004), Fechteler and Eisert (2009) have built up their own nonformal sequence to respect some specific characteristics used in their segmentation algorithms. Others, such as Lavoie et al. (1999) and Zhang et al. (2002) have preferred to use sequences based on the De Bruijn sequences.

The De Bruijn sequence first introduced by Etzion (1988) is based on the finite fields theory in mathematics. The sequences, also called pseudo-random sequences, are generated using shift registers detailed by MacWilliams and Sloane (1976). Their length n is calculated by $n = a^m - 1$, where a is the number of symbols of the sequence (2 in a binary sequence) and has to be a prime number. And m is the degree of the primitive polynomial. For example, if $a = 2$ and $m = 3$, the generated sequence is as follows:

1 0 0 1 0 1 1

If a sequence based on non-prime numbers (like 4) is needed, a combination of numbers can be used to obtain the desired result. For example, a sequence based on four symbols is generated by a binary sequence, where numbers are combined 2 by 2. These sequences have good properties (MacWilliams and Sloane 1976); in particular,

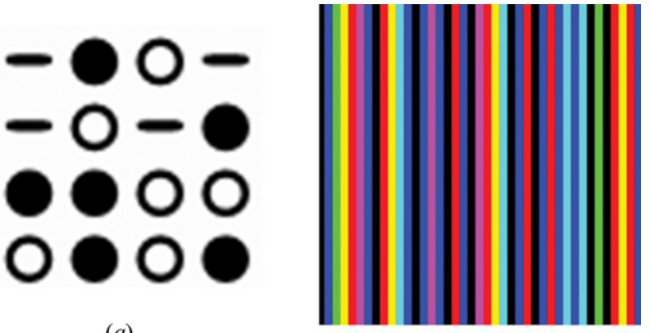
The autocorrelation of a sequence of a length n is given by the following:

$$\begin{cases} \rho(0) = 1 \\ \rho(i) = \frac{1}{n} \text{ for } 0 < i < n \end{cases} \tag{1}$$

The window property: every window of length m is repeated exactly once along the entire sequence. In the sequence above, every window of $m = 3$ elements is unique.

The first property is described by MacWilliams and Sloane (1976) and is the most important as it explains the pseudo-random designation. However, as occlusions may occur in image processing, the most important property in structured light is the window property. It guarantees that every codeword is unique and that the match is unique.

2.4.2. Along one or two axes. Some authors (Albitar et al. 2007; Collewet et al. 2006) have suggested patterns based on the De Bruijn sequence arranged in M-Arrays (also called the perfect maps, see, for example, Figure 1(*a*)).



(*a*) (*b*)

Figure 1. Presentation of the (*a*) two axes (perfect maps) versus the (b) one axis strategy. (*a*) Part of the perfect map proposed by Albitar et al. (2007), they use three symbols: the line, the circle and the filled circle; and (*b*) Pattern made of colored parallel lines proposed by Zhang et al. (2002) (color figure available online).

However, as reminded by Beumier et al. (2004) all of the 3-D information is included in epipolar planes or the horizontal plane if it is assumed the camera and the pattern source are aligned in the latter plane. The former solution has been adopted by many authors (Lavoie et al. 1999; Zhang et al. 2002; Fechteler and Eisert 2009; Beumier 2004, see Figure 1(*b*). These patterns have the great advantage that the resolution is maximized along the vertical axis compared to the M-Array patterns. The use of lines is also better for the horizontal resolution, as lines can be thinner (except for techniques based on the thickness of the lines, such as bar codes). Their weakness is that it should be robust enough to allow the connection of lines split by occlusions.

2.4.3. Defining the right symbols. Several strategies were suggested to define the right symbol; in particular, several authors have chosen the use of colors to distinguish elements (see, for example, Lavoie et al. 1999; Zhang et al. 2002). Unfortunately the use of an SLM requires a monocromatic pattern; colored lines are, therefore, not adapted. Beumier (2004) and earlier Maruyama and Abe (1993) have proposed the use of dashed lines. The latter has proposed the use of dashed lines coded thanks to cuts randomly placed along the lines. The weakness of this method is that occlusions may prevent the identification of the line. Beumier has used the offset related to the previous line to code identify the line (see Figure 2). He used black and white lines to locally detect missing lines. It also allows to double the sequence.

2.4.4. Discussions. The pattern proposed by Albitar et al. (2007) has the required properties for the SLM; however, its density is low (27×29). To solve this issue, the related work presented here shows that the use of a pattern made of lines permits to maximize the information along the vertical axis. Previous pattern made of lines do not fulfil the requirements for an SLM projection. A new pattern needs to be designed to solve these issues.

First, concerning the symbols, Maruyama and Abe (1993) and Beumier (2004) have both proposed the used of dashed lines, which is compatible with our application. The issue that still needs to be solved is the symmetry of the pattern for the SLM projection. Second, concerning the coding strategy, the authors usually prefer the use of a De Bruijn sequence for their robustness. However, despite its good properties, it does not guarantee the required symmetry or the hamming distance between

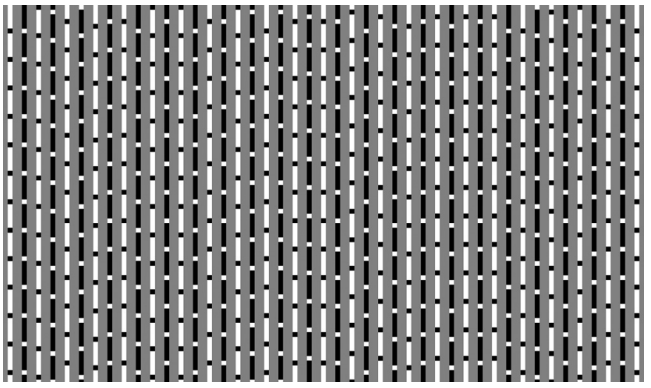


Figure 2. Pattern made of dashed lines coded according to their offset proposed by Beumier (2004).

two consecutive symbols preferred for the robustness of the segmentation. The following section introduces the proposed pattern solving the issues mentioned here.

3. PATTERN CONSTRUCTION

3.1. Requirements

The following constraints were taken into account in pattern design:

- The pattern should allow the reconstruction of the observed surfaces in 3-D with the best resolution. The number of symbols should be maximized by reducing their size and the distance between them. In other words, the density of the pattern has to be maximized.
- The number of elements should be as high as possible in order to reduce the length of the codewords used for the matching. However, the identification is more robust with a low number of elements. A compromise has to be made to deal with these two criteria.
- The pattern is projected by a spatial light modulator.

3.2. Proposed Pattern

3.2.1. Definition of the symbols. Here, is proposed to use a pattern made of dashed and continuous lines (see Figure 3). The proposed coding strategy for dashed lines combines two types of characteristics: the period $p_{i,j}$ of the line and its duty cycle $\frac{a_i}{p_i}$ (see Figure 4). In order to improve the reconstruction of the lines, it is proposed to include continuous lines between two dashed lines: it increases the distance

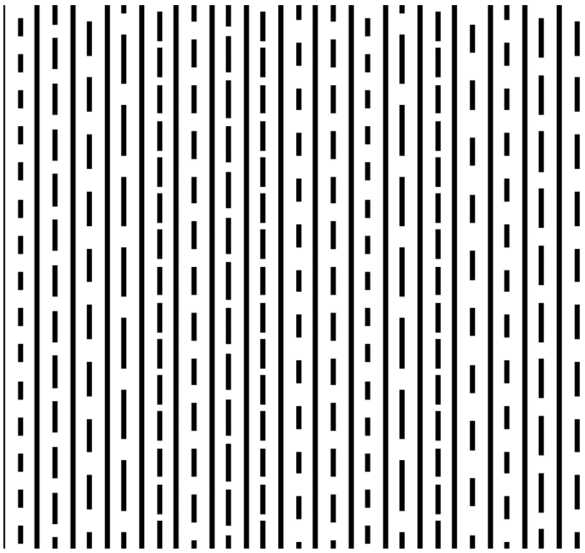


Figure 3. Part of the proposed pattern made of continuous and dashed lines. Dashed lines are coded following their period and duty cycle.

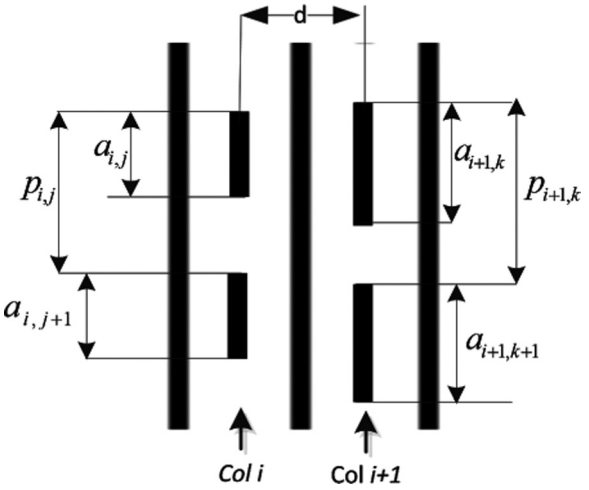


Figure 4. Detail of the pattern. It presents the (1) continuous lines and (2) the segments number j in the line i. Periods and duty cycles are equal for all j, k,.... Segments are repeated with a duty cycle $\frac{a_i}{p_i}$ and a period $p_{i,\,j} = p_i \forall j$.

d (see Figure 4) between dashed lines and reduces the risk to connect two consecutive dashed lines. Therefore, for a pattern of the same size, the sequence can be twice as short. Another advantage, is that these lines can be used for the reconstruction and as they are continuous, they provide a maximum density of information.

3.2.2. Definition of the periods. To increase the robustness and get rid of any geometrical deformation, it is better to avoid any absolute measurements. The easiest way is to identify every line related to its neighbor. As the simple difference between the measured periods still depends on the absolute length, the ratio, between the periods is preferred. Therefore, periods were defined according to a geometric progression: $p = r^n$, where p is the period, r is the progression ratio, and n is $0 \leq n \leq 3$. The parameter r was set to 1.25 to provide a minimum of 25% between two periods. The four generated periods are as follows:

$$1.25^0 = 1$$
$$1.25^1 = 1.25$$
$$1.25^2 = 1.56$$
$$1.25^3 = 1.95$$

During the segmentation process, the decoded periods P_i (see Figure 4) are different, as they are based on the ratio of two consecutive lines periods and are determined as follows:

$$\begin{cases} P_i = P'_i \text{ if } P'_i \geq 0 \\ P_i = (4 - P'_i) \text{ if } P'_i < 0 \end{cases} \qquad \text{with: } P'_i = \log_r(p_i/p_{i-1}), \tag{2}$$

where p_i is the period of the line number i, and P_i is the decoded period (based on the ratio between two measured periods) which is, therefore, different from the original period (based on absolute dimensions).

3.2.3. Definition of the duty cycles. The duty cycle is defined as $\frac{a_i}{p_i}$, as illustrated in Figure 4. A compromise has to be made between the space between segments (which contains no information) and the segmentation robustness. They were defined as follows:

$$0.5\,0.6\,0.7\,0.8$$

3.2.4. Construction of the sequence. As detailed in the previous section, four different periods and four different duty cycles were chosen. Therefore, 16 different symbols exist in the pattern. By combining two lines for the identification, a pseudo-random sequence would theoretically allow to have a $16^2 - 1 = 255$ digits sequence. However, even though the pseudo-random sequences have amazing properties, they do not guarantee other properties recommended for the image processing and the use of an SLM. First, they are not symmetric, which is required to increase the transmitted power; and second, as every codeword appears exactly once, codewords made of two identical items are included in the sequence. This latter property is not recommended since the risk to confuse two contiguous dashed lines is higher. The generation of the sequence followed the process below.

- A binary De Bruijn sequence was generated using the primitive polynomial (as detailed in a previous section) of degree 8.

$$x^8 + x^6 + x^5 + x + 1 \tag{3}$$

- The 255-bits sequence was then converted into an 63-number sequence by combining four consecutive bits.
- For every number of the sequence I, a period and a duty cycle were defined as follows.

$$\text{Period} = (I \bmod 4)^{1.25}$$

$$\text{Duty cycle} = \begin{cases} 0.5 \text{ if } I < 4 \\ 0.6 \text{ if } 4 \leq I < 8 \\ 0.7 \text{ if } 8 \leq I < 12 \\ 0.8 \text{ if } 12 \leq I < 16 \end{cases}$$

- The sequence was then made symmetric; its length was doubled to 126.
- As illustrated in Table 1, the decoded sequence (which is not required to be symmetric) is calculated by the algorithm.

To guarantee the discrimination of the elements between the lines as mentioned above, two checks were made on both the original (important for the robustness of the image processing) and the decoded sequence (important for the robustness of the codeword identification).

Table 1. Comparison of the original and the decoded sequence

Original	Sequence	7	15	8	6	3	8	10
	$\log_{1.25}(p_i)$	3	3	0	2	3	0	2
	Duty cycle	0.6	0.8	0.7	0.6	0.5	0.7	0.7
Decoded	Sequence	6	12	11	6	3	11	10
	P_i (see Eq. 2)	N/A	0	3	2	3	3	2
	Duty cycle	0.6	0.8	0.7	0.6	0.5	0.7	0.7

- Introducing the symmetry also introduced identical pairs in the decoded sequence. Therefore, for every pair of numbers, the following are checked.
  - Is it unique?
  - The hamming distance = 2 for consecutive pairs.
- On the original sequence, it was checked whether three consecutive dashed lines have the same length. As illustrated in Figure 4, lengths $a_{i,j}$ and $a_{i+1,k}\,\forall j, k$ should be as different as possible to improve the discrimination of the lines. The use of the combination of the period p_i and of the duty cycle could lead to segments of the same length (calculated as period × duty cycle). For example, a period of four and a duty cycle of 0.6 has the same segment length as a period of 3 and a duty cycle of 0.8.

The original symmetric sequence is given here:

2 11 12 14 7 2 4 8 15 3 11 7 15 2 8 13 2 15 4 1 5
4 9 1 8 15 4 10 9 14 2 3 2 14 9 10 4 15 8 1 9 4
5 1 4 15 2 13 8 2 15 7 11 3 15 8 4 2 7 14 12 11 2

The decoded sequence is also given:

6 3 11 10 15 5 14 0 9 4 0 12 8 5 2 11 7 11 15 7 12
13 3 4 1 9 15 2 1 11 4 7 5 8 1 3 14 9 3 7 0 13 15
4 13 9 5 9 1 6 11 12 0 4 8 3 12 6 15 9 10 1 5 10 13

3.2.5. Verifications. Hamming distance. Figure 5 illustrates the hamming distance of every pair of elements across the entire decoded sequence. Red (= central line) is used where the hamming distance equals 0, orange (randomly placed dots) where it equals 1 (10.8% of the pairs, excluding the central line), and green where it equals 2 (89.2%). The figure also illustrates that all of the contiguous pairs have a hamming distance equal to 2. It also proves that the window property of the De Bruijn sequence is conserved.

Autocorrelation. The autocorrelation of the decoded sequence did not keep the property described by Equation (1). However, the result is always below 0.2 (see Figure 6), which is good enough to be used as a discriminating criterion.

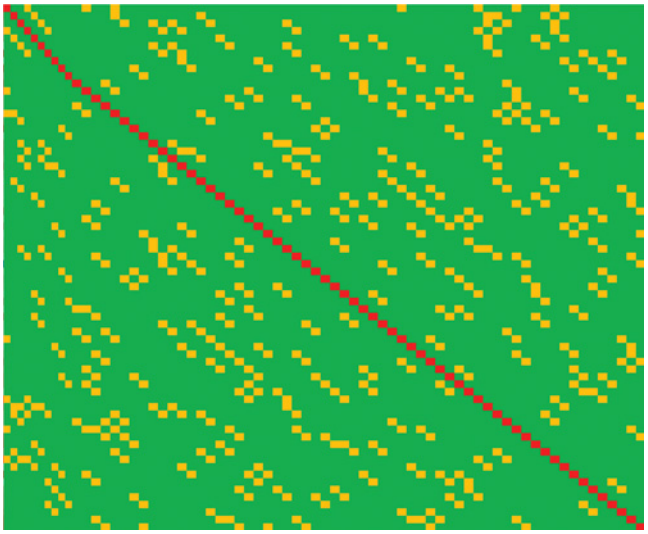


Figure 5. Matrix of the hamming distance between all the pairs of numbers along the decoded sequence (color figure available online).

Length of contiguous lines. The segments' length difference of every contiguous lines were checked. The average difference in length is 40.0% (related to the previous line), and the minimum is 2.5% occurring twice. The same check was made for the neighbor in $i \pm 2$. The result is an average difference of 41.1% and a minimum of 5%.

Resolution. The pattern has 127 parallel lines, which can provide a resolution of 127 points along the horizontal axis and theoretically maximized along the vertical axis. The total surface covered by the pattern is the average of 64% (average of duty cycle) and of 100% (continuous lines) = 82%. Considering that two lines are spaced of 10 pixels, the density achieves up to $\frac{82/\%}{10} = 8.2\%$.

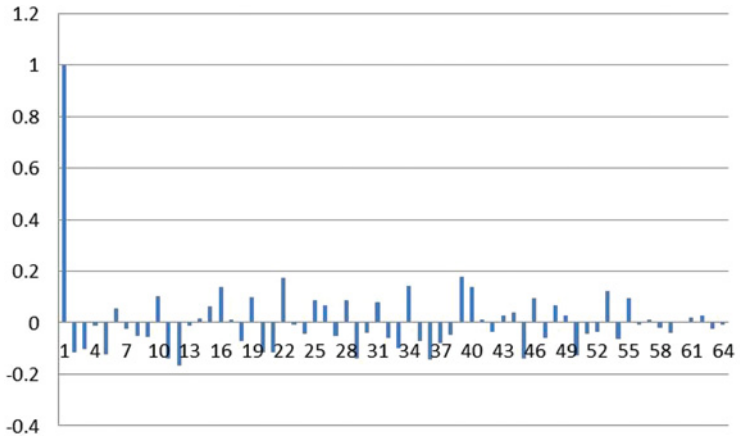


Figure 6. Autocorrelation of the decoded sequence (color figure available online).

Table 2. Comparison of the pattern versus patterns of Beumier (2004) and Albitar et al. (2007)

	Proposed pattern	Albitar	Beumier
Density	8.2% = 39,360 pts	0.2% = 960 pts	8.3% = 39,840 pts
Horizontal res.	127 pts	27 pts	70 pts
Vertical res.	82% = 492 pts	29 pts	83% = 498 pts
Central symmetry	Yes	Yes	No
Coding strategy	De Bruijn-based	M-Array	Un-formal
Codeword length	2×2 for cont. lines	9	3
Number of symbols	16	3	6

Resolutions are illustrated based on an 800×600 pixels image

3.2.6. Comparison with other patterns. Table 2 compares the proposed pattern with the prior art patterns of Beumier (2004) and Albitar et al. (2007). The former, because the principle is very close to the one proposed here; and the latter because it has been designed to be used in the same conditions (SLM and endoscopy). Both of them are also coded based on the neighborhood strategy. For a dashed lines based patterns, the vertical resolution is the relative vertical filling applied to an 800×600 pixels image. The density for Beumier is calculated as 1 dot out of 6 vertically (= duty cycle of $\frac{5}{6}$) for a line every 10 pixels horizontally. The density is $\frac{5}{6} \times \frac{1}{10} = 8.3\%$. The latter could reach a higher density by increasing the duty cycle (for example, to $\frac{9}{10}$). Table 2 shows that the pattern proposed here has a higher density than Albitar's and is similar to Beumier's, a higher horizontal resolution, and is adapted to the SLM projection.

4. RESULTS

4.1. Simulation Setup

The pattern was tested in 3-DSMAX, to exactly quantify the error made on the reconstruction without the influence of the calibration errors. However, the model does not take into account several parameters, such as lens distortions, specular reflection, diffusion, or the blur due to the depth of field. The modelized camera (see Figure 7) has a field of view (FOV) of 40° horizontally and of 30.5° vertically. It corresponds to a lens of 50 mm. A light source modelizes the laser and the SLM with a FOV of 36°. The latter projects a pattern made of 127 lines. Dashed lines are made of 17 to 33 segments depending on their period. The generated image is made of 800×600 pixels. The model was built to modelize an endoscopic application. In the latter, the distance between the pattern source and the camera is typically of $5 - 7$ mm, and the working depth usually remains below 100 mm. In the model, the projector and the camera are horizontally spaced 6 mm. First, the pattern was projected onto a flat plane located at a known distance (75 mm) and on a sphere centred in (0, 0, 100 mm) with a radius of 40 mm. Second, the pattern was projected onto a 3-D model of a stomach in order to evaluate the quality of the reconstruction.

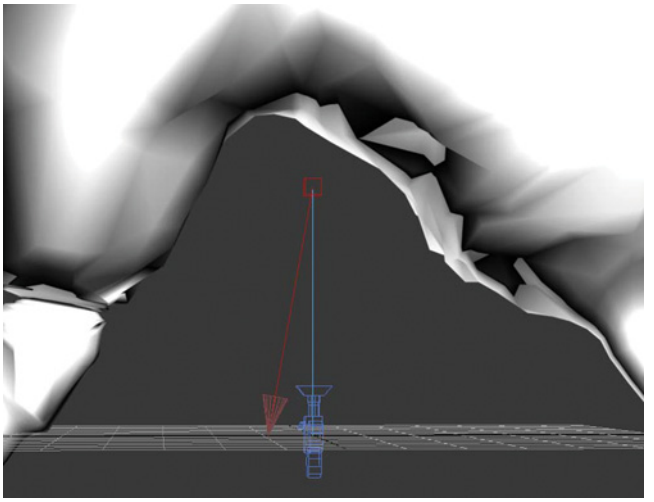


Figure 7. Simulation setup implemented in 3-DSMax (color figure available online).

4.2. Segmentation

The segmentation of the pattern was made using the following algorithms implemented under OpenCV for python (version 2.1.0) using a threshold and a Canny (or labelize) filter for the detection of the segments. The detected period and duty cycles of the segments are shown in Figure 8. The lines were then constructed as follows.

- The K-Means algorithm is used to detect the closest neighbors of every segment. The best is the one for which $\frac{V_i \cdot V_j}{h_i/h_j}$ is maximum, where V_i is the vector director of the segment i, and h_i is the height of the segment i. If $h_i/h_j < 1$, i and j are swapped.
- Reconstruction of the lines and determination of the period and of the duty cycle.
- The reference pattern and the decoded pattern are matched; then, the triangulation is used to find the 3-D positions.

4.3. Quantitative Evaluation of the Error

The first step of the quantitative analysis is to determine whether the detection of the duty cycle and of the period is robust. During the analysis of Figure 9(b), 1,170 segments were identified.

The period was determined with an average error of 1.5 pixel or a relative accuracy of 5.5% (for a period of 27 pixels). The duty cycle was determined with an average error of 2.48%. The lines shown in Figure 8 indicate the period and duty cycle of the reconnected lines. It appears that the risk of confusion is low.

It can be generalized from Kytö et al. (2011), that the depth relative error of 3-D reconstruction in triangulation is given by the following:

$$\left|\frac{\delta z}{z}\right| = \left|(\delta(\tan(\alpha)) + \delta(\tan(\beta)))\frac{z}{d}\right|. \tag{4}$$

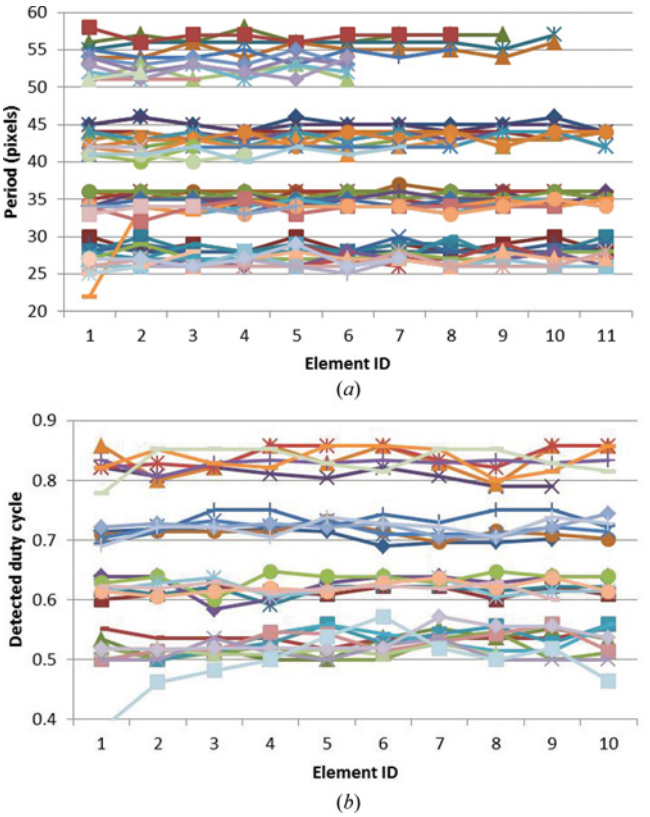


Figure 8. Detection of the properties of the segments to identify them. The dots are connected according to their line. The gathered lines show the robustness of the detection. (*a*) Detection of the period of the elements $ID = j, k, \ldots \forall j, k$ of the line $i, i+1, \ldots$; and (*b*) detection of the duty cycle of the elements $ID = j, k, \ldots \forall j, k$ of the line $i, i+1, \ldots$ (color figure available online).

The relative error based on this equation is plotted in Figure 10.
Where (see Figure 11):

- $\frac{\delta z}{z}$ is the relative error at working depth z.
- d is the distance between the camera and the projector.
- α and β are, respectively, the angle between the observed point and the camera's straight axis and the same for the projector.
- $\delta \tan(\alpha) = \frac{\delta x_p}{f_p}$, where x_p is the pixel position of the object and f_p is the focal length expressed in pixels.

In this case, the camera has a field of view FOV $= 40°$, and the image has a width of 800 pixels. The focal length is calculated as: $f_p = \frac{400\,\text{pixels}}{\tan(20\,\text{deg})} = 1099\,\text{pixels}$. $\delta\beta = 0$ as the projector is supposed to be perfectly characterized. A lower bound of the relative error on the position due to an error of 1 pixel is given by $\frac{\delta z}{z} = \frac{1}{1099} \times \frac{75}{6} = 1.14\%$.

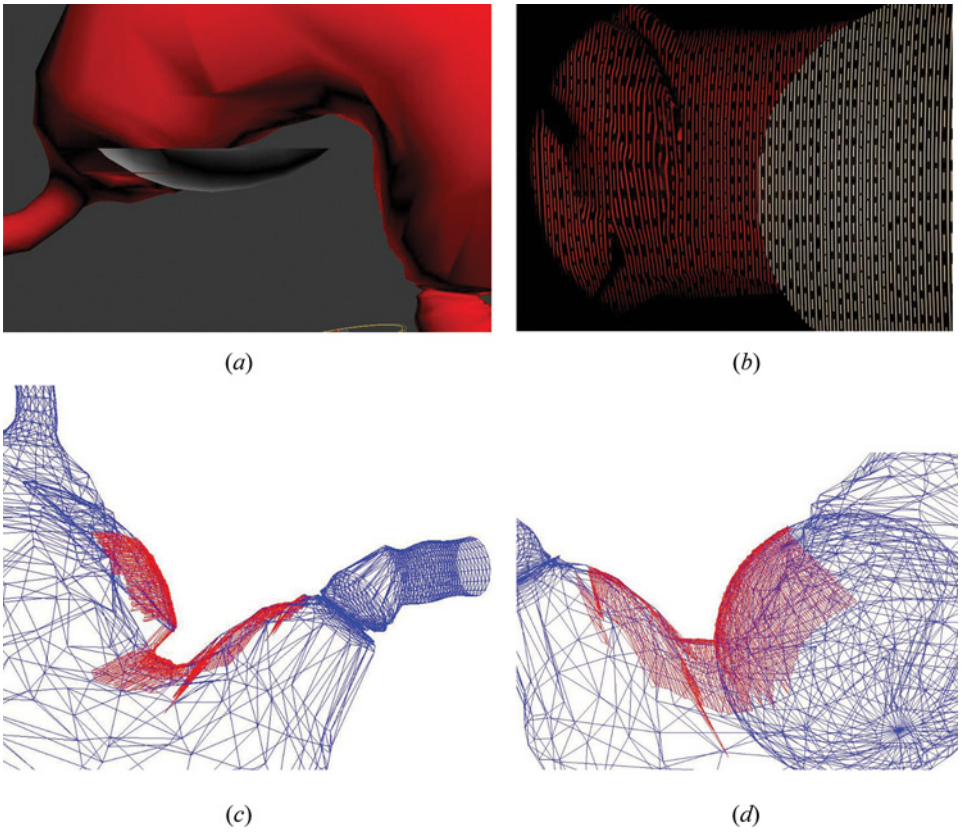


Figure 9. 3-D reconstruction of a sphere and a stomach. The blue model is the original one, and the red one is reconstructed. (*a*) Original model in 3DSMax, (*b*) Camera Acquisition, (*c*) Reconstructed model (in red) and original (in blue), (*d*) Reconstructed model of a sphere and a stomach (color figure available online).

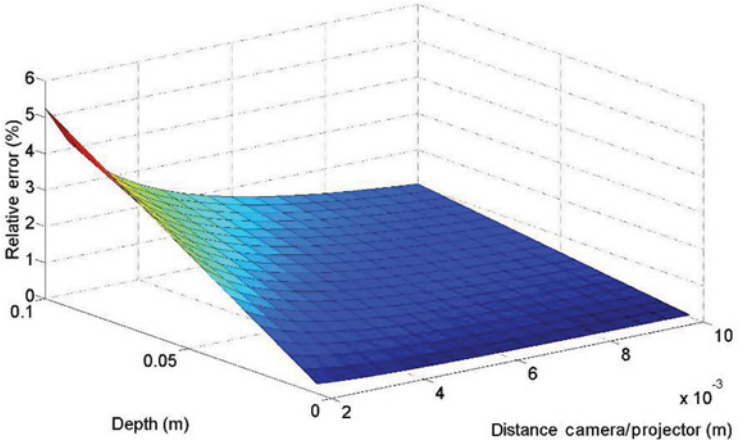


Figure 10. Curve of the 3-D reconstruction relative theorical error (color figure available online).

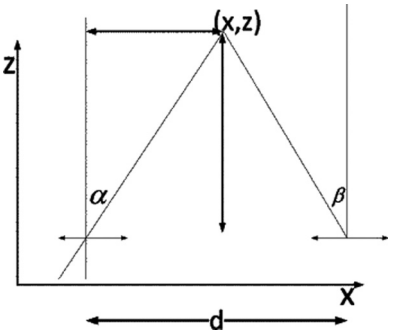


Figure 11. Illustration of the parameters used for the calculation of the error.

Table 3. Reconstruction error on a sphere and a plane

		Absolute	Relative
Plane	Maximum error	0.61 mm	0.8%
	Average error	0.18 mm	0.25%
	Standard deviation	0.227 mm	0.3%
Sphere	Maximum error	0.39 mm	0.99%
	Average error	0.11 mm	0.28%
	Standard deviation	0.108 mm	0.27%

The analysis of the reconstruction presented in Table 3 shows the experimental error compared to the theoretical error. It proves that the reconstruction provided a supbixel accuracy.

4.4. Qualitative Evaluation of the Quality of the Reconstruction

The pattern has then been projected on a 3-D model of a stomach partially occluded by an object. The reconstruction is presented in Figure 11(*c*). The figure also shows that the occlusions were properly managed. Figures 11(*c*) and 11(*d*) also show that a high number of vertices were generated.

5. CONCLUSION AND FURTHER WORK

In this article, a high density pattern made of dashed and continuous lines has been introduced. The lines were coded thanks to specific periods and duty cycles allowing the use of a monochromatic pattern and all its associated advantages. The combination of the period and the duty cycle provides a robust way to identify 16 different types of lines. The introduction of continuous lines has made the segmentation easier and more robust. The sequence was generated based on a De Bruijn sequence, and arranged to obtain additional important properties required by the SLM and the image processing, such as: 1) symmetry of the pattern, and 2) guarantee that segments of two contiguous lines have different specific lengths.

Table 4. Classification proposed by Salvi et al. (2010) applied on the proposed pattern

Compatible with moving objects	Yes
Number of cameras required	1
Number of axes along the pattern is coded	1
Number of colors used in the pattern	1
Is the pattern periodic?	No
Subpixel accuracy achieved?	Yes
Compatible with colored objects?	Yes

The implementation of this pattern on a 3-DSMax model has shown that a subpixel accuracy was achieved and that the quantitative and qualitative results are good enough for the application. Salvi et al. (2010) have introduced a classification of the patterns using several characteristics. The properties mentioned in Salvi et al. (2010) are applied for the proposed pattern in Table 4.

Compared to previous work in this field, this pattern presents three main advantages. First, compared to Albitar et al. (2007) who had 783 elements in their M-Array pattern, here a pattern with 127 lines along the horizontal axis and a maximum resolution along the vertical axis is proposed. Second, compared to Beumier (2004), Maruyama and Abe (1993), Albitar et al. (2007), the amount of lines required for the identification is reduced to 2. Third, the segmentation is made more robust by the introduction of the continuous lines.

Further work will focus on the implementation of the pattern in a device to validate it in real conditions. Depending on the distortion, reflection, the camera resolution or other parameters, the dimensions of the projected pattern and the size of the segments (height, thickness) will be tuned. This will have a direct implication on the achieved spatial resolution. In case of poor camera resolution, it may also be required to reduce the amount of different periods or duty cycles from 4 to 3 or 2. In the latter case, the sequence will have to be adapted.

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